

9. Data logging, file layout, and capture budget

Save every frame raw (.mc), with one manifest row per frame, so that any analysis can be re-run without re-capturing. The manifest records the read-back robot pose, not only the commanded one.

Folder layout (under Noise Estimation Data/):

```
Characterization_<YYYYMMDD>/  sensor_config.json  registration.json  targets_asbuilt.csv
manifest.csv  session_log.md  environment_log.csv  00_registration/  A_noise/  B_edges/  B_zstep/
C_area/  D_detect/  sentinels/
```

File name: <proc>_<target>_G<gap>_Z<zzzz>_F<field>_P<pose>_f<frame>.mc, for example C_T5_G15_Z0800_F0_P017_f03.mc. The pose index has three digits, four for the optional repeats planned outside the budget; a target without a back plate writes G0; the frame index starts at f00.

Manifest columns, in the order the manifest module writes them: file name; procedure letter; target ID; gap G (mm, 0 for a target without a back plate); station Z (mm); field position; pose index; frame index; random seed; commanded lateral offset (H, V mm); read-back robot pose (6 values: position in mm and a rotation vector in degrees); target pose in the camera frame from the registration (6 values); timestamp; sensor temperature; air temperature; sensor-configuration ID; sub-series label (main, tilt, remount, nominal, jitter, ladder, ramp, field, extended, sentinel, and the optional staircase and sweep); tilt axis and angle; commanded step and visit (B-Z); feature index; then metadata columns such as the achieved field fraction and the sentinel's mount-reference flag.

Capture budget. The estimate below assumes a frame rate of 10 frames/s (FRAME_RATE_HZ) and 3 s per move plus settle (MOVE_AND_SETTLE_TIME_S). Both are assumptions; the VSX3000 frame rate in the chosen trigger mode should replace them. Excluded from the budget: target swaps, warm-up, mount checks and the registration's own time, the D post check, and every optional capture: the filters-off repeat of A and B, the staircase and lateral-sweep second passes, the open-background variant, and the D reuse option. The plan tool recomputes the table from Section 2 and lists the optional captures separately. Allow two to three working days in total.

Series	Poses	Frames	Robot time (h)
Registration	30	300	0.03
A main + tilt + sentinels	89	7,070	0.3
B-HV edges	520	15,600	0.87
B-Z ladder + ramp	453	4,530	0.5

C main + field	1,160	11,600	1.29
D main trials	2,160	2,160	1.86
D extended (low-point series)	2,880	2,880	2.48
Total	7,292	44,140	7.3

The extended low-point series takes about a third of the robot time. If time is short, 150 trials per configuration still measure the 5% point, to about $\pm 3.5\%$, and roughly halve that row. Check storage before starting: multiply the size of one .mc frame by about 44,000 frames.